

RedHat Certified System Administrator

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1 Linux Distributions

Linux Distributions or distro are complete system build around **linux kernel**

Linux distribution are differentiated by what sit around the linux kernel
Different linux distro use different linux package manager to **install, update and remove** software from a centralized controlled source. For RedHat we will talk about **rpm**.

Almost anything in a linux distribution is **Open Source**.

All linux distros share core linux tools like databases and programming languages.

RedHat follow the **File Hierarchy Standard (FHS)** for default file position.

RedHat releases new major versions every **three year**. Leap help you with a smooth transition from one version to the next one. Version support come in three phases:

- Full support for the first 5 years
- Maintenance support for the next 5 years
- Extended update support for selected security patches

2 Introduction to the shell

Commands are sunned on a shell and the default shell is bash.

Basic structure of a command is:

command |options| arguments|

Option begin with a dash (–) or a double dash (––)

Options influence how the command run, while argument are the things you want your command to work on

3 Documentation

man pages are already on the system and tell you how commands work.

To access a man page for a command type: man |command|

Use `man 5 jcommand` to access the man page for the command configuration file

4 Linux Directory

Linux system have a root directory `/`, inside the root live predefined directory used to store linux file:

- `home`: used to store account user specific files
- `etc`: extended text configuration store the configuration file for your system
- `var`: is where you find the variable files
- `usr`: store user specific program
- `lib64`: is the directory used to store common library
- `tmp`: store the temporary file cleaned up periodically
- `root`: is the directory for the root user

5 Basic File Management

6 Using Image Mode with bootc

Point server to images with complete OS app and configuration.

Produce brand new image for running application.

Use the bootc image with `systemd` command to deploy and run local application.

7 Image builder

Golden images are perfect template to build upon your custom image.

They are a clean and consistent starting point for any server or VM.

RHEL Image builder is the cli tool use to manage and launch image on your local environment.

8 Vulnerability Management